

Multiple Choice

1. **Answer:** D

Options: A) Ammonium hydroxide; B) Sulfuric acid; C) Acetic acid; D) Potassium hydrogen phthalate

Note: Correct option: D - Potassium hydrogen phthalate

2. **Answer:** A

Options: A) Blackbody radiation curves; B) Emission spectra of diatomic molecules; C) Electron diffraction patterns; D) Temperature dependence of reaction rates

Note: Correct option: A - Blackbody radiation curves

3. **Answer:** D

Options: A) BaH₂; B) BaOH; C) Ba; D) BaO

Note: Correct option: D - BaO

4. **Answer:** B

Options: A) 0.0471 Hz; B) 21.2 Hz; C) 42.4 Hz; D) 66.7 Hz

Note: Correct option: B - 21.2 Hz

5. **Answer:** B

Options: A) Ti³⁺; B) Cr³⁺; C) Fe³⁺; D) Zn²⁺

Note: Correct option: B - Cr³⁺

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'SCl₂'.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '2-methylpentane'.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '3.5 R'.

Long Form Response

11. **Answer:** The atomic radii of the lanthanide elements increase across the period from La to Lu.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** The total entropy of the system plus surroundings increases.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key